

DSC40B:
Theoretical Foundations of Data
Science II

Lecture 9: *Hashing and Hash table*

Instructor: Yusu Wang

Recall: (Dynamic) set operations

- ▶ Imagine you are maintaining a database indexed by some keys (real values), and you hope to support the following operations:
 - ▶ Search
 - ▶ Maximum
 - ▶ Minimum
 - ▶ Successor
 - ▶ Predecessor
- ▶ Insert
- ▶ Delete
- ▶ Extract-Max
- ▶ Increase-key

Using balanced BST, all these operations can be done in $O(\lg n)$ time



Dictionary operations

- ▶ Given a universe of elements U
 - ▶ these elements may not be numbers
- ▶ Need to store some keys
- ▶ Need to perform the following operations for keys
 - ▶ Search (**membership queries**)
 - ▶ Insert
 - ▶ Delete

We can use balanced BST for this purpose if keys can be compared.

But we can have even lighter weight data structure to handle these



Today

- ▶ Hashing in general
- ▶ Hash table
 - ▶ For dictionary operations



Part A:
The idea of Hashing



Hashing

- ▶ An important idea used commonly in practice
- ▶ Many uses:
 - ▶ Fast queries on a large data set.
 - ▶ Verifying message integrity.
 - ▶ Identify if file has changed in version control.



Hash function

- ▶ Mathematically, a hash function is simply a function
 - ▶ $f: U \rightarrow X$ from one set to another
- ▶ To make it useful, in practice,
 - ▶ It often maps some potentially large or complex object to a much smaller and simpler “fingerprint” or “signature”
 - ▶ One also wants the mapping to be “uniform” and cause few “collisions”.
 - ▶ Note: a hash function needs to be deterministic!
 - ▶ Hashing the same object twice, we should get the same answer



Some examples

- ▶ A cryptographic hash function:
 - ▶ maps data of arbitrary size into an often fixed-sized output of much smaller size
 - ▶ e.g, MD5: maps it to a 128-bit value
 - ▶ hard to “reverse engineer” input from hash.
 - ▶ two similar input could and should lead to very different hash values



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 - ▶ two similar input could and should lead to very different hash values

```
▶ > echo "My name is Justin" | md5  
▶ a741d8524a853cf83ca21eabf8cea190  
▶ > echo "My name is Justin!" | md5  
▶ f11eed2391bbd0a5a2355397c089fafd
```

```
▶ > md5 slides.pdf  
▶ e3fd4370fda30ceb978390004e07b9df
```



A cryptographic hash function

- ▶ Why?

- ▶ I release a piece of software.
- ▶ I host it on Google Drive.
- ▶ Someone (Google, US Gov., etc.) decides to insert extra code into software to spy on users.
- ▶ A user has no way of knowing.



A cryptographic hash function

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- ▶ Someone (Google, US Gov., etc.) decides to insert extra code into software to spy on users.
- ▶ A user has no way of knowing.

▶ What do I do?

- ▶ I release a piece of software and **publish the hash**
- ▶ I host it on Google Drive.
- ▶ Someone inserts extra code into software to spy on users.
- ▶ You download the software and hash it. If hash is different, you know the file has been changed!



Some examples

- ▶ **Want to place images into 100 bins.**
 - ▶ How do we decide which bin an image goes into?
 - ▶ Hash function!
 - ▶ Takes in an image.
 - ▶ Outputs a number in $\{1, 2, \dots, 100\}$



Part B: Hash Table



Dictionary operations

- ▶ Given a universe of elements U
- ▶ Need to store some keys
- ▶ Need to perform the following operations for keys
 - ▶ Insert
 - ▶ Search
 - ▶ Delete
- ▶ In other words, the key operation is **membership queries** (search), but also allows **dynamic updates** (insert, delete).



Approach 0

- ▶ Use an array to organize all the keys
 - ▶ We could pre-sort the array.
 - ▶ Preprocessing time: $\Theta(n \log n)$
 - ▶ Search: $\Theta(\log n)$
 - ▶ Insert / Delete: $\Theta(n)$



Approach 1

- ▶ Organize all keys in a doubly-linked list

- ▶ Pre-processing: **None**

- ▶ Insert: $\Theta(1)$

- ▶ Delete

$\Theta(1) \rightarrow$ assume we know the ~~pos~~^{pointer} of the element to be deleted.

- ▶ Search:

$\Theta(n)$

Approach 2

- ▶ Organize all keys in a balanced BST
- ▶ Search: $\Theta(h) = \Theta(\lg n)$
- ▶ Insert: $\Theta(\lg n)$
- ▶ Delete: $\Theta(\lg n)$



Approach 3

- ▶ Direct address tables (DAT)



Example: Query for age

- ▶ Is there any student whose age is 17 ?

PID	Name	Age
A1843	Wan	24
A8293	Deveron	22
A9821	Vinod	41
A8172	Aleix	17
A2882	Kayden	4
A1829	Raghu	51
A9772	Cui	48
⋮	⋮	⋮



Approach 3

- ▶ **Direct address tables (DAT)**
- ▶ Suppose we know that all the keys we ever care are from 0 to $N = 150$
 - ▶ Like in the case when we query for ages,
- ▶ Open an array A of size N
 - ▶ If a given age z is in, set $A[z] \geq 1$, and 0 otherwise
 - ▶ Given a query age, say 22, simply return $A[22]$
- ▶ Search: $\Theta(1)$
- ▶ Insert: $\Theta(1)$
- ▶ Delete: 



What's the problem of DAT approach?

- ▶ DAT (directed address tables) very efficient
- ▶ However
 - ▶ Require the keys to be integers
 - ▶ Require the keys to be a fixed range
 - ▶ Require a size as large as this range – which makes it hard to use for a huge universe (of keys)



How about checking for names?

- ▶ Did the name Smith appear in the student database?

PID	Name	Age
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Hash Table:

A way to use DAT idea, but allowing to use much more general universe through the “hash” map.



Use Hash Table!

- ▶ U : universe

- ▶ $T[0 \dots m - 1]$: a hash table of size m

 - ▶ $m \ll |U|$

 - ▶ usually, we choose m to be around the size of data we expect to see

- ▶ Hash functions

 - ▶ $h: U \rightarrow \{0, 1, \dots, m - 1\}$

 - ▶ i.e, h maps each element in the universe to an index in the hash-table

- ▶ $h(k)$ is called the **hash value** of key k .

 - ▶ Given a key k , we will store it in location $h(k)$ of hash table T ,

 - ▶ i.e, store it at $T[h(k)]$

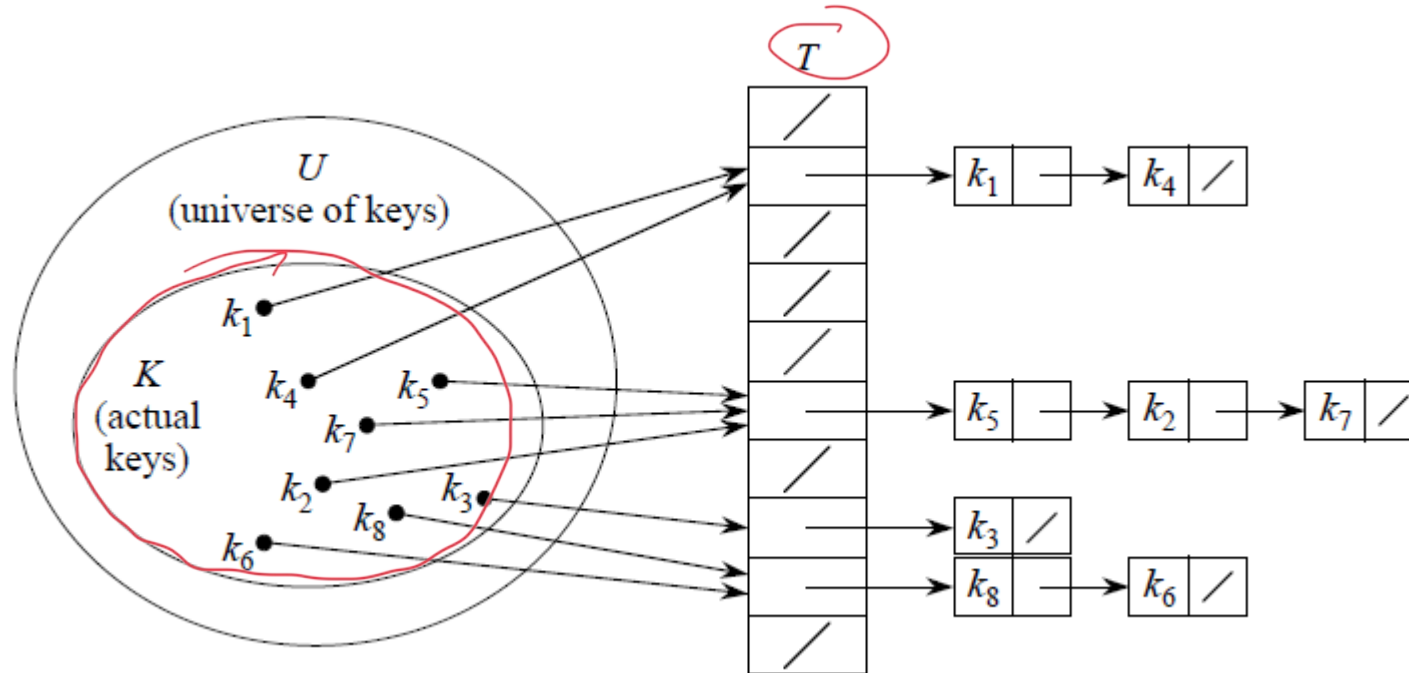


Collision

- ▶ Since the size of hash table is smaller than the universe:
 - ▶ Multiple keys may hash to the same slot.
 - ▶ A **collision** happens when $h(x) = h(y)$ for $x \neq y \in U$
- ▶ How to handle collisions?
 - ▶ Chaining
 - ▶ Open addressing
 - ▶ ...



Collision Resolved by Chaining



- ▶ $T[j]$: a pointer to the head of the linked list of all stored elements that hash to j
- ▶ Nil otherwise

A simple example $P = \{22, 35, 3, 9, 10, 44\}$

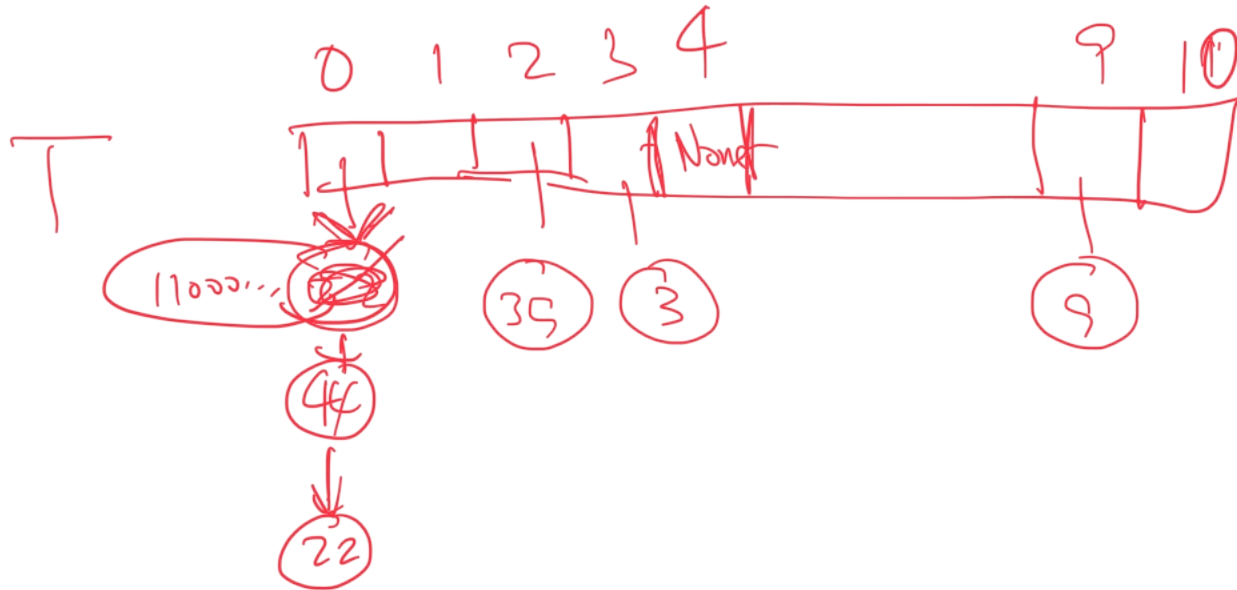
- ▶ U : all positive integers $\mathbb{N}$
- ▶ Hash table $T[0, \dots, 10]$
- ▶ Hash function: $h: \mathbb{N} \rightarrow [0, \dots, 10]$
 - ▶ $h(x) = x \bmod 11$

$$P = \{22, 35, 3, 9, 10, 44\}$$

Is $15 \in P$?

$$h(15) = 4$$

$T_S \approx 22$ in P ?

$$h(22) = 0$$


$$h(22) = 0$$

$$h(35) = 2$$

$$h(3) = 3$$

$$h(a) = a$$

$$h(44) = 0$$

Dictionary operations

▶ Chained-Hash-Insert (T, x)

- ▶ Insert x at the head of list $T[h(x)]$

$$\Theta(1)$$

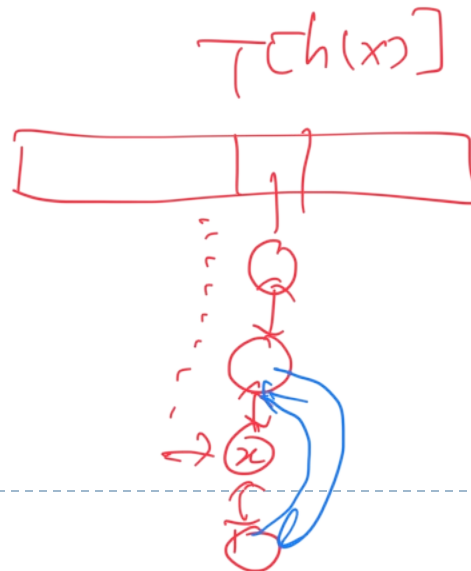
▶ Chained-Hash-Search(T, x)

- ▶ Search for an element with key x in list $T[h(x)]$

$$\Theta(\text{length of } T[h(x)]) = O(n)$$

▶ Chained-Hash-Delete(T, x)

- ▶ Delete x from the list $T[h(x)]$



$$\Theta(\text{length of } T[h(x)] + 1)$$

↓
search time

$$= O(n)$$

Dictionary operations

- ▶ Chained-Hash-Insert(T, x)

- ▶ Insert x at the head of list $T[h(x)]$

$O(1)$

- ▶ Chained-Hash-Search(T, x)

- ▶ Search for an element with key x in list $T[h(x)]$

$O(\text{length}(T[h(x)]))$

- ▶ Chained-Hash-Delete(T, x)

- ▶ Delete x from the list $T[h(x)]$

$O(\text{length}(T[h(x)]))$



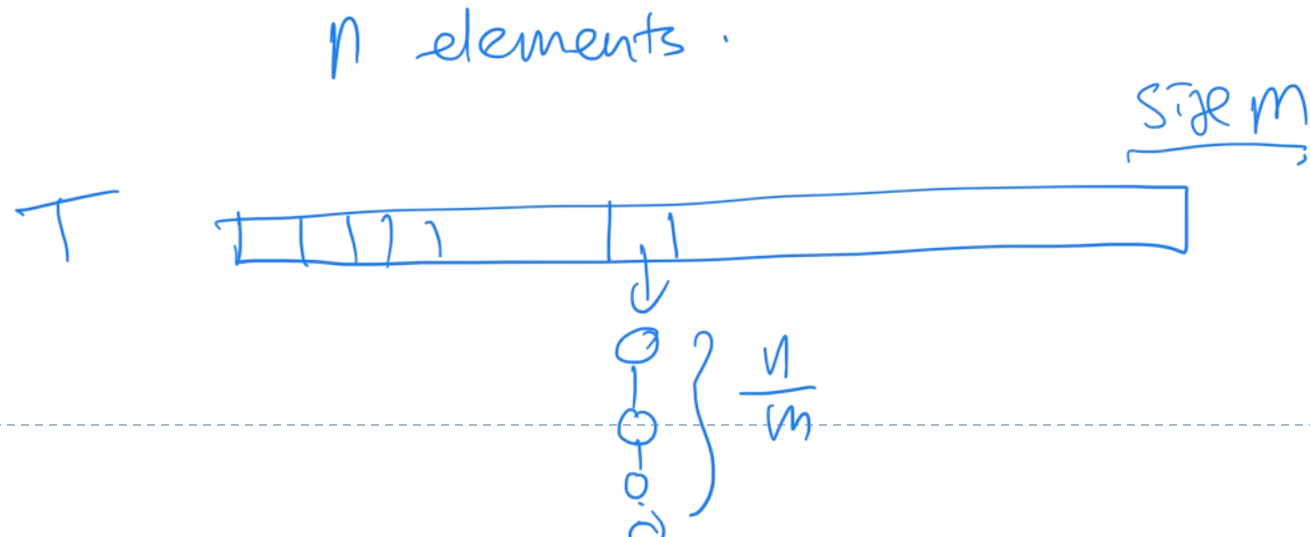
Good Hash Function

- ▶ Performance of Hash table operations
 - ▶ depend on the number of elements hashed to each slot in hash table.
- ▶ Intuitively,
 - ▶ A good hash function should spread elements into the hash table uniformly
 - ▶ If there are n elements in the input data and m slots
 - ▶ then ideally there should be $\frac{n}{m}$ elements in each hash table slot.

Search:

worst case, $\Theta(n)$

best of worst case $\Theta(\frac{n}{m})$



Average case analysis

- ▶ n : # elements in the table
- ▶ m : size of table (# slots in the table)
- ▶ **Load factor:**
 - ▶ $\alpha = \frac{n}{m}$: average number of elements per linked list
 - ▶ Intuitively the optimal time needed
- ▶ Individual operation can be slow ($\Theta(n)$ worst case time)
- ▶ *Under certain assumption of the distribution of keys*, analyze expected performance.



Average case analysis

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- ▶ Individual operation can be slow ($\Theta(n)$ worst case time)
- ▶ Under certain assumption of the distribution of keys, analyze expected performance. In particular, for a search query:
 - ▶ Time to find the correct bin: $\rightarrow \Theta(1)$ (compute hash value)
 - ▶ Expected number of elements in this bin: $\rightarrow \Theta\left(\frac{n}{m}\right)$
 - ▶ Time to perform linear search in this bin: $\rightarrow \Theta\left(\frac{n}{m}\right)$
 - ▶ Total: $\Theta\left(1 + \frac{n}{m}\right)$

Average case analysis

- ▶ n : # elements in the table
- ▶ m : size of table (# slots in the table)
- ▶ **Load factor:**
 - ▶ $\alpha = \frac{n}{m}$: average number of elements per linked list
 - ▶ Intuitively the optimal time needed
- ▶ Individual operation can be slow ($\Theta(n)$ worst case time)
- ▶ *Under certain assumption of the distribution of keys*, analyze expected performance. In particular, for a **search query**:
 - ▶ Time to find the correct bin: $\Theta(1)$
 - ▶ Expected number of elements in this bin: n/m
 - ▶ Time to perform linear search in this bin: $\Theta(n/m)$
 - ▶ Total: $\Theta(1 + n/m)$



Expected running time

- ▶ Under Simple uniform hashing assumption

- ▶ Search:

- ▶ Expected time: $ET(n) = \Theta(1 + \frac{n}{m})$

- ▶ (worst case time $T(n) = \Theta(n)$)

- ▶ Insert:

- ▶ $T(n) = \Theta(1)$

- ▶ Delete:

- ▶ Expected time: $ET(n) = \Theta(1 + \frac{n}{m})$

- ▶ (worst case time $T(n) = \Theta(n)$)

Key message:
if the load factor $\alpha = \frac{n}{m} = \Theta(1)$
then $\Theta(1)$ expected time for these operations!

In summary

- How fast can we query / insert with these data structures?

	Query	Insert
Unsorted linked list	$O(n)$	$O(1)$
Unsorted array	$O(n)$	$O(1)$
Sorted array	$O(\log n)$	$O(n)$
BST	$O(\log n)$	$O(\log n)$
Hash Table (<u>expected time</u>)	$\Theta(1)$	$\Theta(1)$

when the assumptions are satisfied

Growing the Hash Table

- ▶ Insertions / search take $\Theta(1)$ expected time under the simple uniform hashing assumption, and that $\frac{n}{m} \leq c$ for some constant c
(m is circled in blue with an arrow pointing to "size of hash table")
- ▶ We need to ensure that $n \leq c \cdot m$
- ▶ As we add more elements (thus n increases), we might need to increase m
 - ▶ -- need resizing strategies in hash table



Hashing in Python

- ▶ **dict** and **set** implement hash tables
 - ▶ **set** : stores only a set of values
 - ▶ **dict** : stores a set of (key, value) pairs, and query is on “key”

- ▶ Querying is done using **in**:

```
>>> # make a set
```

```
>>> L = {3, 6, -2, 1, 7, 12}
```

```
>>> 1 in L # Theta(1)
```

```
False
```

```
>>> 7 in L # Theta(1)
```

```
True
```



Part C:

Some examples of using hash tables
and downsides of hash tables



Recall the movie problem

- ▶ The Movie problem
 - ▶ Input: Given a list of length of movies available, stored in array *movies*, and a flight duration D
 - ▶ Output: Return two movies whose total length = D ; **None** otherwise.



Recall

- ▶ The naïve algorithm solves it in $\Theta(n^2)$ time
- ▶ But earlier, we mentioned we can do better by
 - ▶ First sorting the array of movie lengths
 - ▶ Then check for each movie x , whether there exists one whose length is $D - \text{length}(x)$
 - ▶ Worst case time complexity $T(n) = \Theta(n \lg n)$



New approach via Hashing

{120, 80, 83, 105, ...}

- Use Hash table, and frame this as a membership query problem

```
def optimize_entertainment_hash(times, D):  
    hash_table = dict()  
    for i, time in enumerate(times):  
        hash_table[time] = i  
  
    for i, time in enumerate(times):  
        target = D - time  
        if target in hash_table:  
            return i, hash_table[target]  
    return None
```

$D = 220$
210

New approach via Hashing

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Expected time?
 $\Theta(n)$



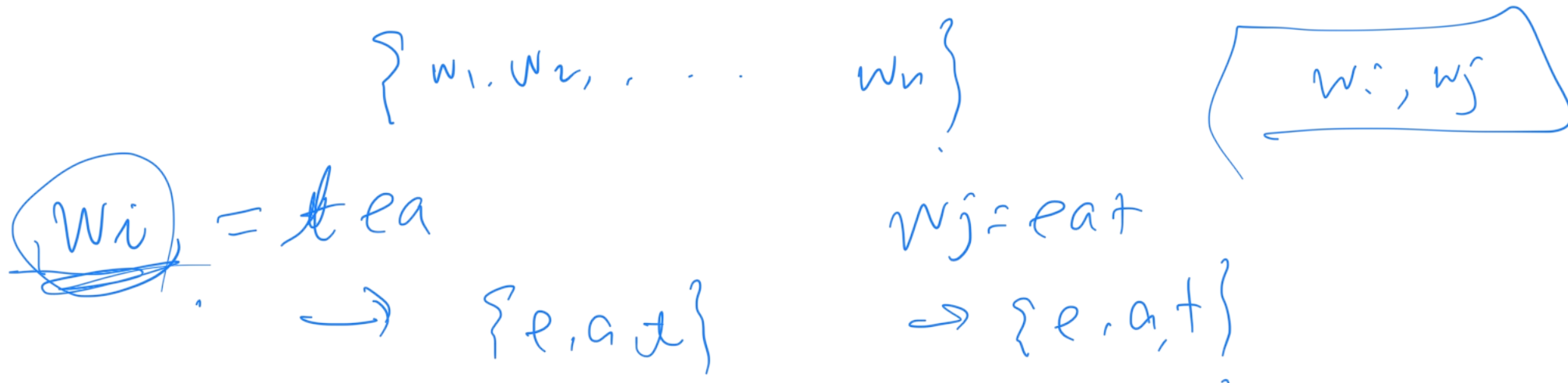
Another example

► Anagrams

- Two strings w_1 and w_2 are **anagrams** if the letters of w_1 can be permuted to make w_2 .
 - E.g, “**e**at” and “**t**ea”, or “**l**isten” and “**s**ilent”

► The *Anagrams problem*

- Given a collection of n strings, determine if any two of them are anagrams.



Using Hash table

► Observation:

- two strings are anagrams if their sorted lists are equal
 - `sorted(w_1) == sorted(w_2)`

```
def any_anagrams(words):  
    seen = set()  
    for word in words:  
        w = sorted(word)  
        if w in seen:  
            return True  
        else:  
            seen.add(w)  
    return False
```

Expected time?
 $\Theta(n)$

Hashing Downsides

- ▶ Only support dictionary queries
 - ▶ i.e, membership queries + insert / delete
 - ▶ Example 1: cannot be used to query for the two movies whose total time is **closest** to D
 - ▶ Example 2: cannot be used for performing a range query in a list of numbers
 - ▶ Say report the number of numbers fall within range $[a, b]$
 - ▶ Cannot be used for min / max / median / order statistics etc
 - ▶ BST wins when “order” type of query is important



Hashing Downsides

- ▶ Only support dictionary queries
 - ▶ i.e, membership queries + insert / delete
- ▶ No locality: similar items map to very different bins
 - ▶ Necessarily so for the performance of hashing in most cases!
 - ▶ But, in practice, we often query similar objects continuously
 - ▶ May result in many cache misses, slow



Summary

▶ Hashing

- ▶ Very useful idea
- ▶ Generates a small signature for a potentially large complex object

▶ Hash Table

- ▶ Very practical data structure for **dictionary operations**
 - ▶ Very efficient for these operations, can be constant expected time if the load factor $\alpha = \frac{n}{m} = \Theta(1)$
- ▶ Especially when the number of keys necessary is much smaller than the size of universe where input objects could come from!
- ▶ In practice, need to choose hash functions properly
 - ▶ There exist intelligent hashing schemes



FIN

